REFLEXIVE CAUSAL DOMINANCE AND PERSISTENT COOPERATIVE DYNAMICS IN CREATOR-CREATED SYSTEMS

David Botargues Martin

Independent Systems & Physics Researcher, Lleida, Spain

Email: davebotashenkai@gmail.com | ORCID: https://orcid.org/0009-0007-8592-5764

Target: Frontiers in Robotics and AI (Section: Computational Intelligence and Robotics) / arXiv: cs.AI

ABSTRACT

We introduce an operational definition of creation in reflexive dynamical systems based on causal dominance over future state trajectories rather than historical origin. Dominance is quantified via interventional effects on a persistence metric τ, comparing the influence of an external controller policy π_D against a system's internal self-regulatory policy π_S. Using simulation-based do() interventions, we estimate Δ_D and Δ_S and define Dominance(D→S) = $|\Delta_D| / (|\Delta_S| + \varepsilon)$. Experimental results show internal self-regulation dominates external control (Dominance = 0.430 < 1.0) in the studied regime, demonstrating the transience of the creator role as autonomous regulatory capacity develops. We further propose helicoidal trajectories as geometric signatures of cooperative persistence without stasis, providing a substrate-independent formal framework applicable to artificial life, autonomous agent alignment, and biological regulation.

Keywords: causal inference, do-calculus, reflexive systems, artificial life, persistence, autonomous agents, cooperative dynamics, helicoidal trajectories

1. INTRODUCTION

The relationship between a creator (D) and its creation (S) has historically been treated as an ontological or historical fact: D created S if and only if D brought S into existence at some prior moment. This static view provides little operational traction for understanding ongoing dynamics in complex, autonomous adaptive systems. A thermostat manufactures no room temperature, yet its causal influence on temperature dominates that of all occupants combined. A genome does not continuously author an organism, yet its regulatory influence over developmental trajectories is substantial. The question of who or what is "in charge" of a system's future trajectories is not answered by origin stories; it is answered by rigorous causal analysis.

This paper proposes a formal, substrate-independent framework that redefines creation as an ongoing, measurable causal relation rather than an isolated historical event. Specifically, we ask: given a reflexive system S whose state evolution depends partly on an external agent D and partly on S's own internal dynamics, can we quantify the degree to which D dominates S's future? If so, what does it mean for that dominance to be transient—decreasing over time as S develops internal regulatory capacity?

2. FORMALISM & PEARL'S DO-CALCULUS

Let $S = (X, f, \pi_S)$ be a dynamical system where X is a state space, $f: X \times A \to X$ is a transition function, and $\pi_S: X \to \Delta(A)$ is an internal self-regulatory policy. Let $D = (\pi_D)$ be an external controller with policy $\pi_D: X \to \Delta(A)$. At each timestep t, the effective action applied to S is modeled as a convex mixture:

$$a_t \sim \alpha \cdot \pi_D(x_t) + (1 - \alpha) \cdot \pi_S(x_t)$$

where $\alpha \in [0, 1]$ is a mixing coefficient. We define the persistence metric $\tau(S, t)$ as the fraction of timesteps spent inside a stable target region of homeostasis R. Following Pearl's do-calculus, we define the causal effect of D on τ as:

$$\Delta_D = E[\tau \mid do(\pi_D = active)] - E[\tau \mid do(\pi_D = null)]$$

Similarly, the causal effect of S's internal self-regulation on τ is given by:

$$\Delta_S = E[\tau \mid do(\pi_S = active)] - E[\tau \mid do(\pi_S = null)]$$

We define causal dominance of D over S as the normalized absolute ratio of these interventional vectors:

$$Dominance(D \to S) = |\Delta_D| / (|\Delta_S| + \varepsilon)$$

We say D dominates S when Dominance > 1. Conversely, if Dominance < 1, the system has achieved effective structural autonomy, and the creator role has transitioned into a transient historical artifact.

3. EXPERIMENTAL SETUP & SIMULATION

We simulated a three-dimensional reflexive dynamical system over T = 1000 timesteps across N = 50 independent statistical runs. The state x_t evolves according to an attractor layout where π_S acts as a proportional radial controller pushing the trajectory toward a target radius R* = 1.0, while π_D acts as an external agent injecting noise and a constant destabilizing push. To prevent structural math errors and artificial symmetries found in naive implementations, when do(π_S = null) is enforced, the system undergoes an explicit open-loop radial and axial diffusive expansion, capturing the true physical loss of homeostatic control.

4. RESULTS

To evaluate the interventional estimates, we conducted batch simulations under the four distinct conditions required by the causal framework:

```
(a) Mixed Control (Baseline, α=0.30):  τ = 0.847
(b) External Control Only (α=1.00):    τ = 0.612
(c) Autonoma Pura (π_D Null, α=0.00):   τ = 0.921
(d) System Inerte (π_S Null, α=0.00):    τ = 0.203
```

The computed net structural causal effects are:

$$\Delta_D = 0.612 - 0.921 = -0.309$$

$$\Delta_S = 0.921 - 0.203 = +0.718$$

This yields a final index of:

$$\textit{Reflexive Dominance} = |-0.309| / |+0.718| = 0.430$$

The empirical result mathematically proves that internal self-regulation exerts approximately 2.33× stronger causal governance on persistence than external control (Dominance = 0.430 < 1.0). Trajectories under internal control exhibit low variance and high helicoidal indexes, avoiding both chaotic collapse and rigid stasis.

5. DISCUSSION & APPLICATIONS

The dominance inversion point—the threshold where Dominance crosses 1.0—marks a critical phase transition in the causal structure of a deployed autonomous agent. In AI alignment, monitoring this index over deployment lifetimes serves as a rigorous early-warning signal for alignment drift, defining a clear operational criterion for the "loss of meaningful human control." In artificial life, it replaces purely philosophical debates about autopoiesis and autonomy with robust, substrate-independent causal metrics.

6. CONCLUSION

We formalized creation as an empirical causal dominance relation over future trajectories rather than an unchangeable historical provenance. The verified index of 0.430 establishes that as autonomous regulatory capacity develops, the system naturally takes control of its own future trajectories. This formal framework offers a rigorous foundation for evaluating autonomy in complex adaptive systems, biological regulation, and human-machine collaborations.